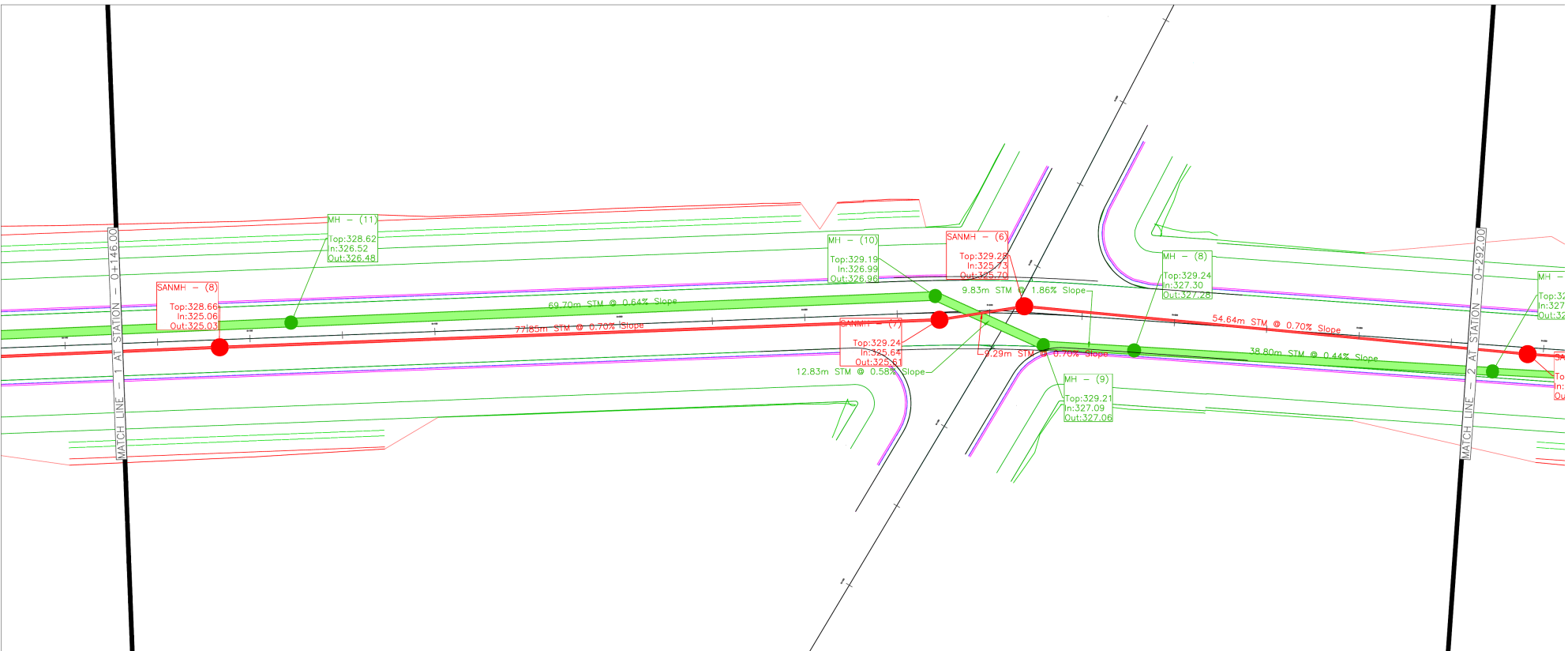


Municipal Road and Sewer Design

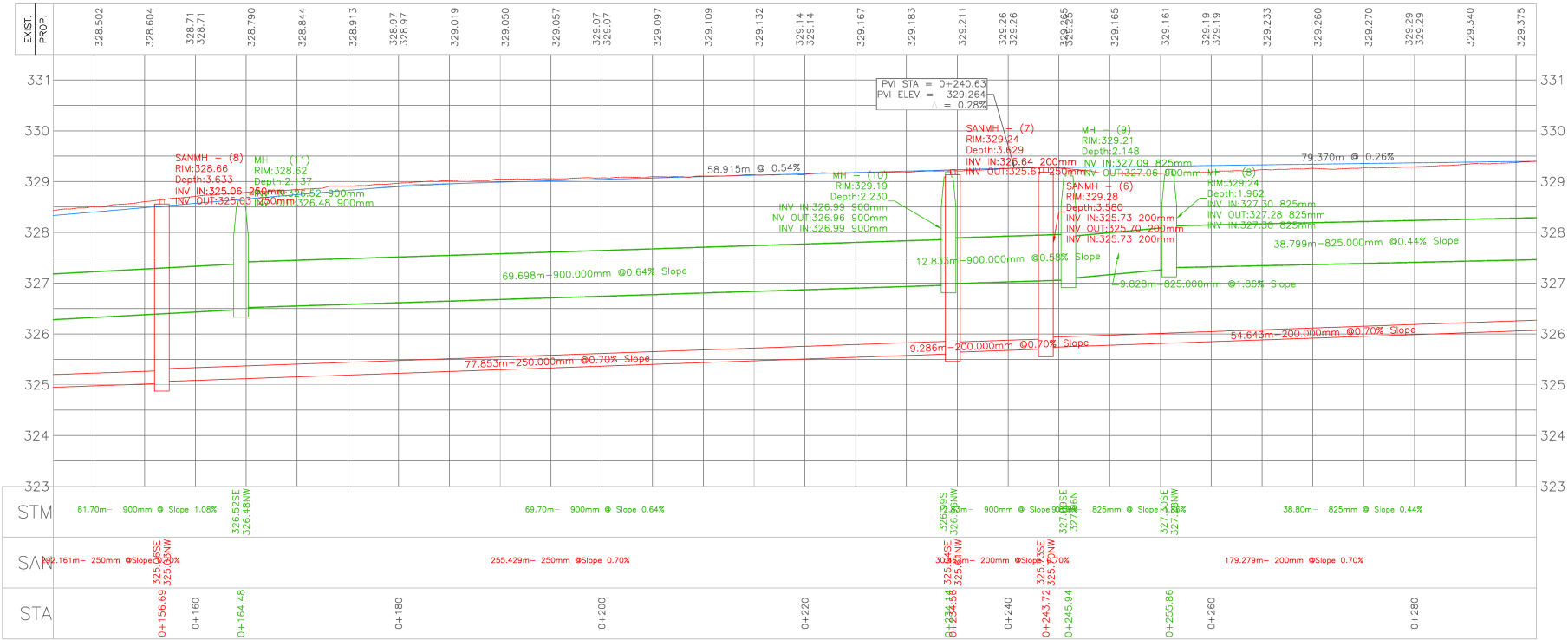
Conceptual plan and profile drawing package

SELECTED CAPABILITIES

- Road corridor and intersection coordination
- Storm and sanitary plan/profile production
- Conceptual GIS/LiDAR existing-condition modelling



PROFILE



4 PLAN/PROFILE EXHIBITS
ROAD / STORM / SANITARY
CONCEPTUAL - NOT FOR CONSTRUCTION

Coordinated municipal road and sewer design

Design intent translated into a controlled Civil 3D model

This independent portfolio uses public GIS and LiDAR-derived information to demonstrate a complete conceptual workflow: existing-ground development, road and intersection modelling, storm and sanitary coordination, and plan/profile sheet production.

PROJECT TYPE Municipal road reconstruction and underground servicing	DESIGN PLATFORM Autodesk Civil 3D with data shortcuts, XREFs and sheet sets
PRIMARY OUTPUT Four coordinated colour plan/profile drawing exhibits	PROJECT STATUS Conceptual portfolio project - not for construction

DELIVERED IN THE MODEL AND DRAWING PACKAGE

01Existing context Terrain and mapping inputs organized into an existing-ground reference.	02Road system Alignment, profile, corridor, intersection and lane-width transition coordination.	03Buried services Storm and sanitary networks displayed in plan and profile with structures and bands.	04Sheet production View frames, match lines, XREF coordination, annotation and colour publication.
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PROFESSIONAL BOUNDARY

The work demonstrates modelling and drawing-production skills. Existing conditions and buried utilities require survey, records review and field verification. The package is not approved, sealed, tendered or construction-ready.

Criteria applied to the conceptual design

Defined controls before detailed modelling

The following criteria were used as portfolio design controls based on applicable local municipal infrastructure requirements. They describe the modelled intent; they do not replace project-specific standards, survey, calculations, approvals or licensed engineering review.

DESIGN CONTROL	MODELLED INTENT	CURRENT EVIDENCE	NEXT QA GATE
Road grades	Geometric controls and vertical transitions	FG profiles and corridor references present	Check criteria and tie-in elevations
Lane drainage	Typical 2.0% crossfall to gutter	Road and offset profile workflow demonstrated	Confirm every region and superelevation case
Lane transition	3.5 m lane transitioning to 3.0 m	Corridor width transition modelled	Review taper geometry and feature-line continuity
Storm cover	Target 1.5 m minimum	Target documented; exported minima include values below target	Rebuild network, resolve exceptions and verify report
Sanitary cover	Target 2.5 m minimum	Intended criterion; independent cover export not available	Export network report and verify every pipe
Pipe geometry	Positive slope, coordinated inverts and structures	Plan/profile networks and bands demonstrated	Check cover, clearances, crossings and connectivity

SENIOR REVIEW POSITION

The cover values are design targets, not completed compliance claims. The current storm export contains exceptions below the target, and an equivalent sanitary verification export was not available. Before any real use, rebuild the networks, resolve exceptions, confirm downstream capacity and hydraulics, and obtain survey and licensed engineering review.

Controlled model-to-sheet production

One source of truth, staged through explicit QA gates

The production approach separates source data, design objects and presentation files so revisions can be traced and sheets can be rebuilt without manual drafting drift.

01	SOURCE DATA Public GIS and LiDAR inputs	QA GATE Check coordinate system, extents, units and data limitations.
02	BASE MODEL EG surface and reference geometry	QA GATE Review surface boundaries, profiles and anomalous elevations.
03	ROAD DESIGN Alignment, FG profile, corridor and intersection	QA GATE Check tie-ins, lane widths, crossfall, targets and regions.
04	PIPE NETWORKS Storm and sanitary pipes and structures	QA GATE Check connectivity, slopes, inverts, cover and crossings.
05	PRODUCTION XREFs, view frames, match lines and profile bands	QA GATE Check ownership, styles, station ranges and annotation.
06	PUBLICATION Sheet set, colour PDF and portfolio wrapper	QA GATE Check plotting, legibility, sheet continuity and final PDF.

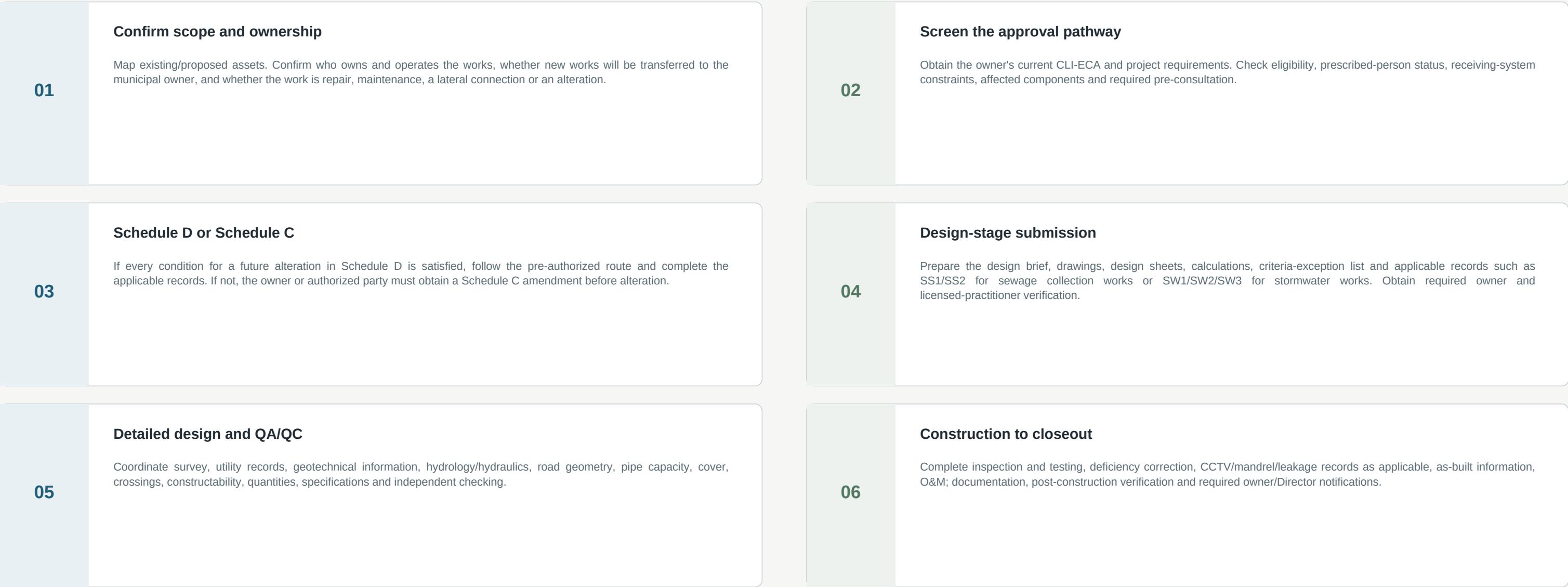
REVISION PRINCIPLE

Revise the owning Civil 3D object or source drawing, synchronize references, rebuild the corridor and networks, then regenerate and visually inspect the sheets.

Ownership, authorization and closeout workflow

Authorization is established before detailed design

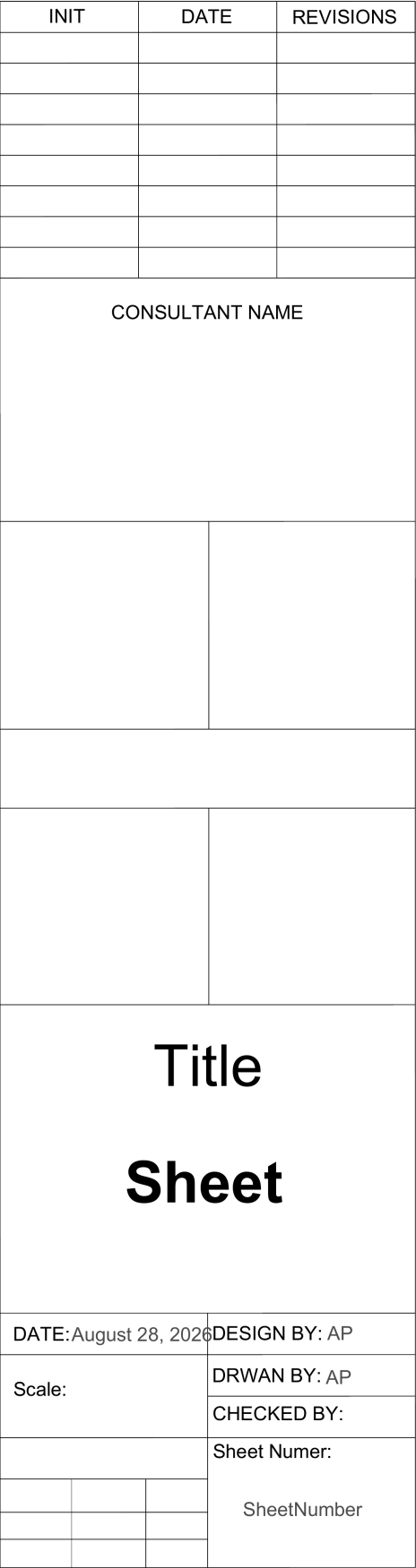
A real project begins by confirming ownership, regulatory scope and the approval pathway. CLI-ECA screening applies to municipal sanitary collection and stormwater works; it does not replace planning, environmental assessment or other permits.

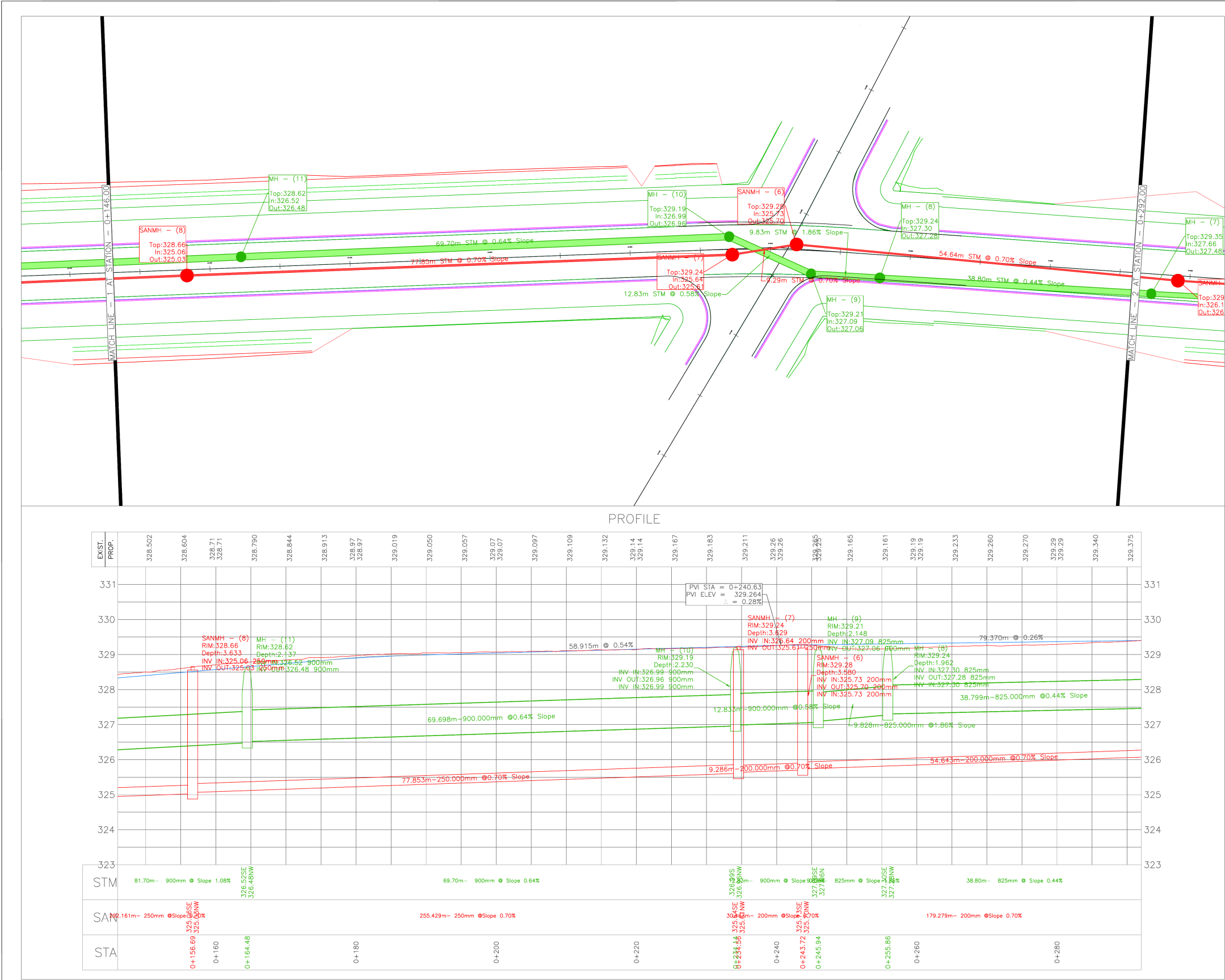


PARALLEL SCREENING

Confirm applicable planning and Environmental Assessment Act obligations, conservation authority and source-protection requirements, utility and road permits, heritage/archaeology, property/easements, traffic staging, erosion and sediment control, and stakeholder consultation.

Regulatory source basis: Ontario MECP municipal CLI-ECA guidance and forms, reviewed September 2026.



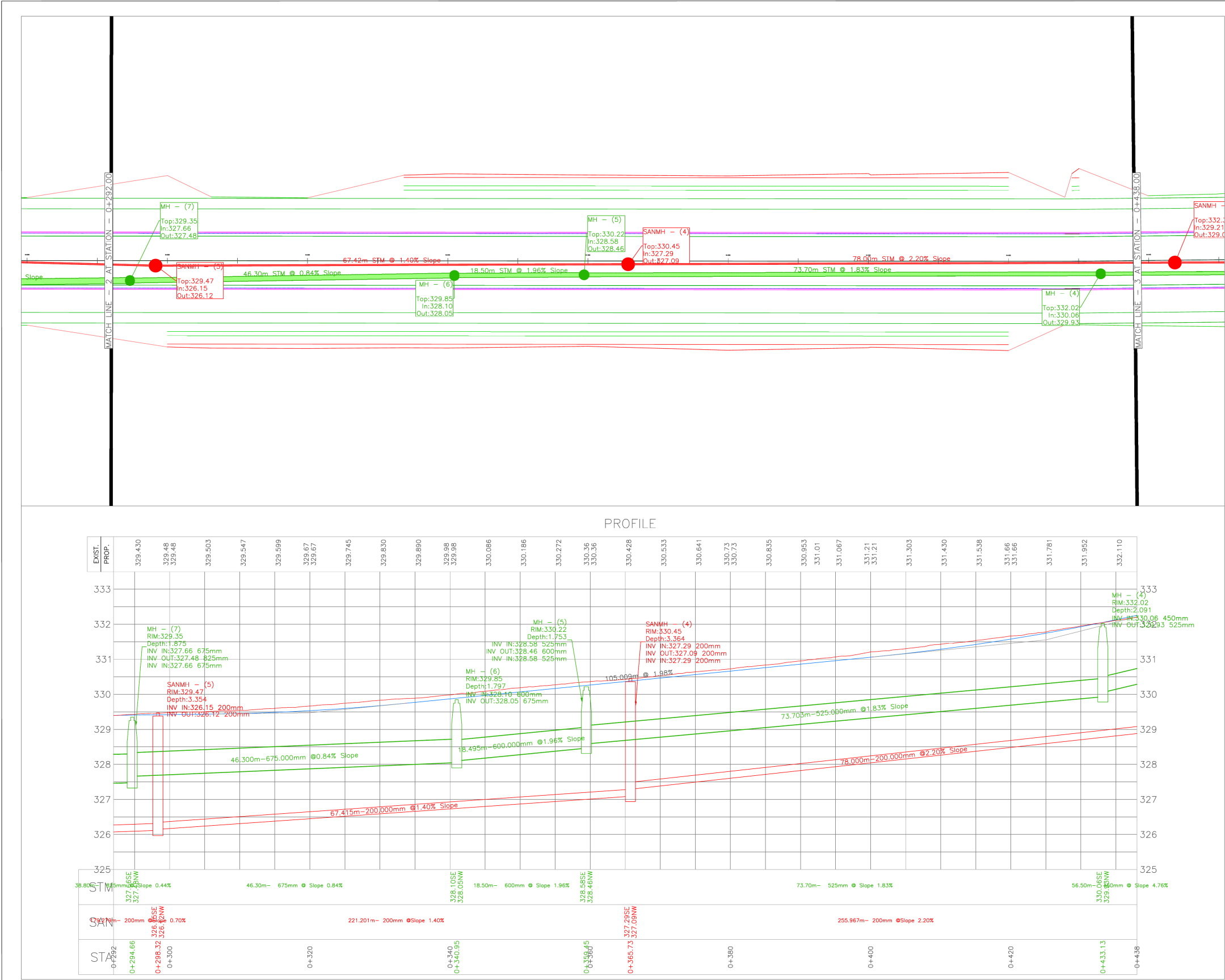


INIT	DATE	REVISIONS

CONSULTANT NAME

Title Sheet

DATE: August 28, 2026	DESIGN BY: AP
Scale:	DRWAN BY: AP
	CHECKED BY:
	Sheet Numer:
	SheetNumber



INIT	DATE	REVISIONS

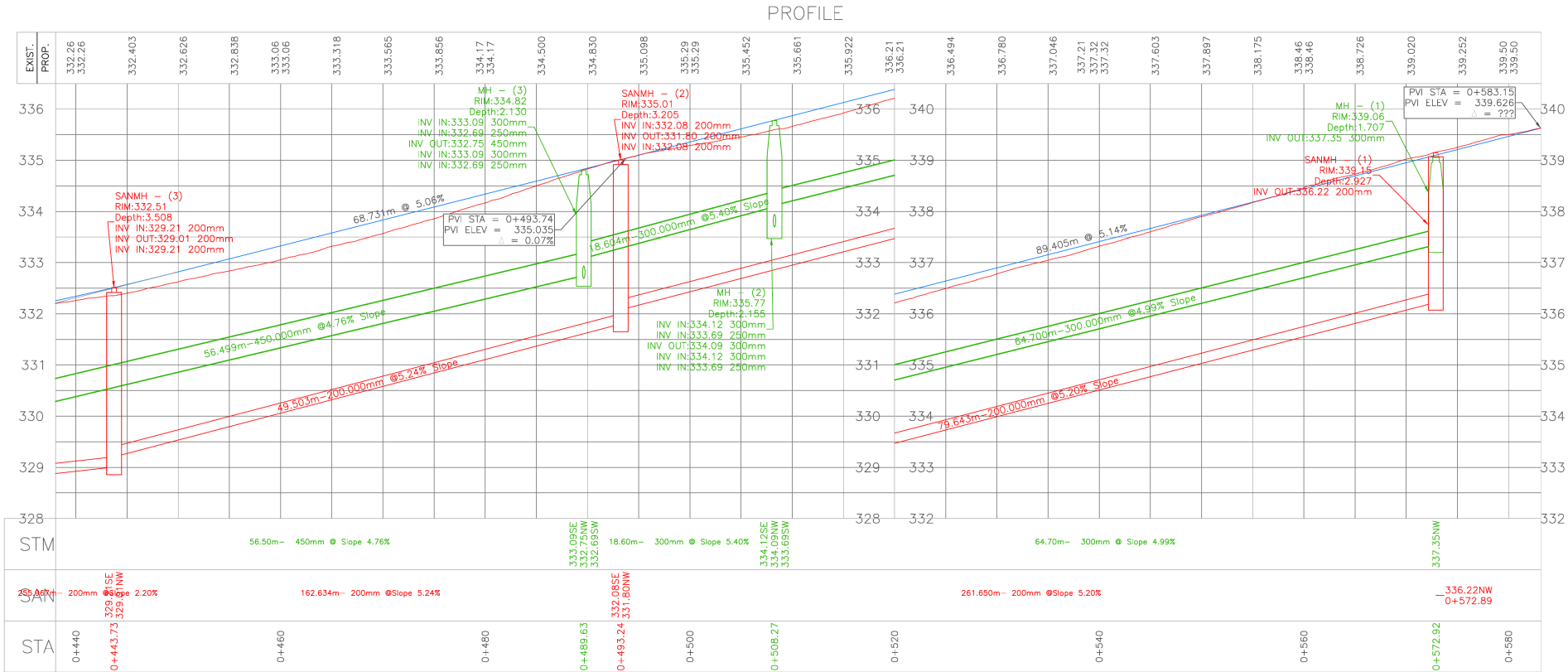
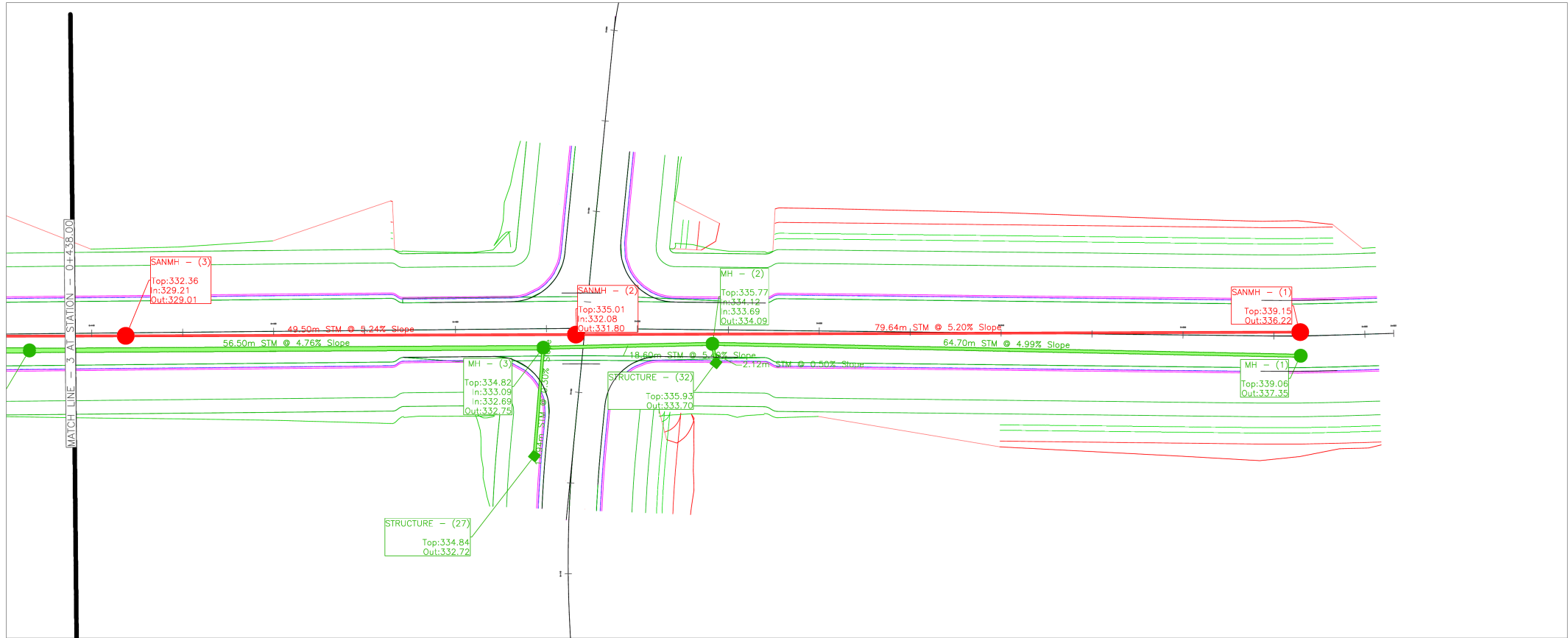
CONSULTANT NAME

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Title
Sheet

DATE: August 28, 2026	DESIGN BY: AP
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	Sheet Numer:
	SheetNumber



INIT	DATE	REVISIONS

CONSULTANT NAME

Title
Sheet

DATE: August 28, 2026 DESIGN BY: AP

Scale: DRWAN BY: AP

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Sheet Numer:

SheetNumber

Demonstrated work and remaining verification

A credible portfolio separates completed work from professional sign-off

The model and sheets demonstrate a substantial Civil 3D production workflow. The items at right remain mandatory before similar work could support a real municipal submission or construction package.

DEMONSTRATED IN THIS PORTFOLIO	REQUIRED BEFORE REAL-WORLD USE
Model organization Existing ground, alignments, profiles, corridor and referenced design objects.	Verified base information Legal survey, subsurface utilities, records review, geotechnical information and field checks.
Road coordination Intersection geometry, lane-width transitions, crown-to-gutter grading intent and tie-ins.	Engineering analysis Hydrology, hydraulics, downstream reserve capacity, I&I, pipe sizing and design exceptions.
Network production Storm and sanitary pipes, structures, plan/profile display, labels and bands.	Independent QA/QC Object connectivity, cover, crossings, constructability, quantities, specifications and checking.
Sheet delivery XREF coordination, view frames, match lines, colour plotting and multi-sheet publication.	Authorization and closeout Owner acceptance, applicable approvals, licensed review, inspection, testing and as-builts.
PORTFOLIO USE Prepared to demonstrate Civil 3D modelling, municipal servicing coordination and drawing-production skills. It is not an issued design, permit submission, tender package, record drawing or construction document.	